

04 — Marketing Website + Dashboard Production-Readiness

Revision: 1 **Parent:** 00_MASTER_INDEX.md **Prerequisites:** A-03 (website decisions), A-10 (hxota.dev root behavior)

Overview

The marketing website (§11.4.190 mandate) does NOT exist — only a scaffold is registered at submodules/website (commit 9abb15e). Both the dashboard and ota-manager need §11.4.170 host-render proof completion for all feature pages.

D-01 [OPERATOR] — Resolve Website Decisions

Already documented in 01_OPERATOR_DECISIONS.md §2 and §9. Must be resolved before D-02.

D-02 [AGENT] — Scaffold Angular 22 SSR + Tailwind v4 + OpenDesign Brand (L)

Source: docs/research/website/00_WEBSITE_DESIGN_AND_BUILD_PLAN.md, docs/research/website/01_SCAFFOLD_READINESS_AND_VERIFICATION.md

What to build:

1. **Use existing submodule:** submodules/website (already at commit 9abb15e in .gitmodules). Check what scaffolding exists:

```
ls submodules/website/  
git -C submodules/website log --oneline -5
```

2. Tech stack (from design plan):

- Angular 22 SSR (server-side rendering for SEO)
- Tailwind v4 (matches ota-manager's Tailwind 4)
- OpenDesign “Helix-green” brand layer (tokens from design-systems/helix-ota/)
- Dark/light theme switcher
- i18n support (English primary, extendable)

3. **If scaffold is bare/empty:** Full `ng new` in the submodule, configure SSR, install Tailwind v4, vendor OpenDesign tokens.

4. **If scaffold has content:** Extend and harden.

5. **Containerized build (§11.4.173):** Add Dockerfile that builds the Angular SSR app in a container. Wire into `scripts/remote_deploy/deploy_website.sh`.

D-03 [AGENT] — Build Content to Locked Spec (M)

Source: docs/research/website/00_WEBSITE_DESIGN_AND_BUILD_PLAN.md

Content spec (LOCKED — do not deviate):

- **Sales email:** `contact@hxota.com` (no other contact method)
- **NO pricing** anywhere on the site
- **Footer:** “Made with ♥ by Helix Development team”
- **Roadmap items:** Clearly marked as “Coming Soon” or “Planned”
- **What to describe:** Only REAL, SHIPPING capabilities:
 - Universal OTA update system
 - Android A/B seamless updates
 - Staged rollout with automatic HALT on failures
 - Multi-device fleet management
 - Secure artifact signing (Ed25519)
 - HTTP/3 + Brotli transport
 - REST API + dashboard + CLI

Pages:

1. **Home** — hero, value proposition, key features
2. **Features** — detailed feature descriptions with diagrams
3. **Architecture** — high-level architecture diagram
4. **Documentation** — links to docs (API, deployment, guides)
5. **Contact** — `contact@hxota.com` only

Do NOT include:

- Pricing or “request a quote” forms
 - User testimonials (none exist)
 - Benchmarks or performance claims (unmeasured on production)
 - Download links (artifacts are behind auth)
 - Login/signup (that’s the console, not the marketing site)
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D-04 [AGENT] — §11.4.190 Proofs: Responsive + SEO + Visual (M)

Source: Constitution §11.4.190

Required proofs:

1. **Responsive:** Host-rendered screenshots across breakpoint × browser-engine matrix:
 - Breakpoints: 320px (mobile), 768px (tablet), 1024px (desktop), 1440px (wide)
 - Engines: Chromium, Firefox, WebKit (Playwright supports all three)
 - Layout oracle: OCR-based verification that key elements are visible and correctly positioned
2. **SEO audit:** Automated audit meeting a score floor:
 - Semantic HTML (proper heading hierarchy, landmarks)
 - Per-page `<title>` + `<meta name="description">`
 - OG/Twitter cards
 - Canonical URLs
 - `schema.org` JSON-LD
 - `robots.txt` + `sitemap.xml`
 - WCAG AA compliance (axe-core audit, 0 critical/serious violations)
 - Core Web Vitals (LCP < 2.5s, FID < 100ms, CLS < 0.1)

3. **Light/dark §11.4.170 pixel proof:** Golden image-diff for EVERY page in light and dark themes.
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D-05 [AGENT] — Dashboard: Complete §11.4.170 Host-Render (M)

Current state: Dashboard has strong coverage (26 goldens, ~13 screen×state × {light,dark}) per PRODUCTION_READINESS_PLAN.md §3.

Gap: Feature pages (after login) need host-render for every screen×state combination.

What to add:

1. Audit which dashboard pages lack host-render goldens:

```
ls dashboard/hostrender/
```

2. Add host-render tests for: Devices list, Device detail, Artifacts, Releases, Deployments, Rollouts, Groups, Telemetry, Webhooks, Branches, Projects, Audit log, Settings.
 3. Each page: light+dark, loaded state, empty state, error state.
 4. Use existing harness pattern from dashboard/hostrender/.
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D-06 [AGENT] — ota-manager: Complete §11.4.170 Host-Render (M)

Current state: ota-manager has only 2 screens rendered (LoginPage). Per PRODUCTION_READINESS_PLAN.md §3.

What to add:

1. Host-render tests for: Login (already done), Dashboard, Devices, Artifacts, Releases, Deployments, Rollouts, Groups, Telemetry, Webhooks, Branches, Projects, Audit log, Settings.
 2. Each screen: light+dark, loaded state, empty state, error state.
 3. Use existing harness pattern from clients/ota-manager/visual/.
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D-07 [AGENT] — Deploy Website + Dashboards (S)

What to do:

- 1. **Website:** Build Angular SSR app, deploy to Firebase hosting (target website) or to the remote host's nginx static serving.
- 2. **Dashboard:** Build Vite SPA, deploy to Firebase hosting (target dashboard) or to remote host.
- 3. **ota-manager:** Build Tauri app or serve web version, deploy to Firebase hosting (target ota-manager) or remote host.
- 4. **Verify:** All three are reachable at their domains, load correctly, no console errors.

Verification Checklist

Step	Action	Expected Result
D-02	Website scaffold builds	<code>npm run build</code> exits 0
D-03	Content matches locked spec	No pricing, correct email, heart footer
D-04a	Responsive screenshots	All breakpoints × engines render correctly
D-04b	SEO audit	Score ≥ 90, WCAG AA 0 violations
D-04c	Light/dark pixel proof	Golden image-diff GOOD/BAD correctly
D-05	Dashboard host-render	All feature pages, all themes, all states
D-06	ota-manager host-render	All feature pages, all themes, all states
D-07	Deployed and reachable	HTTP 200 on all three targets

Honest Boundary (§11.4.6)

The submodules/website scaffold exists at commit 9abb15e but was not inspected this session for content. The design plan (docs/research/website/00_*) was read and is authoritative for content decisions. The website build is blocked on operator decisions A-03 and A-10.